Stock Market Prediction — Nifty 500

Product Requirements Document · Data Analytics Intern Project

Owner	Manager / Mentor	Status	Draft v1.1
Intern	Data Analytics Intern	Date	September 2026

1 · Project Overview

Summary

This project tasks a Data Analytics Intern with building a stock price prediction model for the **Nifty 500 index**, using five years of historical market data sourced from NSE and BSE. The intern will practice the full analytics lifecycle — data collection, cleaning, exploratory analysis, feature engineering, model building (statistical, classical ML, and deep learning), evaluation, and delivery of an interactive Streamlit application — culminating in a working prototype and a findings report.

Objective

Predict the future **target price** of the Nifty 500 index using five years of historical data sourced from NSE and BSE, and communicate actionable insights through an interactive Streamlit dashboard and summary report.

Why This Project

- Builds practical experience with time-series data, financial datasets, and predictive modeling.
- Demonstrates the intern's ability to work through an end-to-end data science workflow, including deep learning and app deployment.
- Produces a reusable analytics pipeline that could later be extended to individual constituent stocks or other indices.

Non-Goals

- Not a production trading algorithm — will not be used for live or automated trading decisions.
- Not intended to guarantee investment returns or serve as financial advice.
- Not focused on intraday / high-frequency data — daily granularity is the default scope.

2 · Goals & Success Metrics

Goal	Metric	Target
Build a clean, reliable dataset	% missing / erroneous data after cleaning	< 2%
Predict target price (Nifty 500)	RMSE / MAE / MAPE vs. naive baseline	Outperform naive / moving-average baseline
Deep learning model performance	LSTM RMSE vs. classical ML models	Competitive with or better than classical models
Deliver clear insights	Stakeholder review rating	Positive sign-off from mentor

Goal	Metric	Target
Learning outcome	Intern demonstrates end-to-end pipeline	Final presentation + documented code

Note: prediction accuracy targets are learning benchmarks, not guarantees — markets are highly noisy and largely efficient, so even strong models may only marginally beat a naive baseline. This should be discussed transparently in the final report.

3 · Scope

In Scope

- **Primary study subject:** Nifty 500 index, using 5 years of historical daily OHLCV data sourced from NSE and BSE.
- **Prediction target:** future target price of the Nifty 500 — a regression problem, not just directional classification.
- Data cleaning and preprocessing.
- Exploratory Data Analysis (EDA): trends, seasonality, volatility, correlations.
- Feature engineering: technical indicators (moving averages, RSI, MACD, Bollinger Bands, volatility, lagged returns).
- Model building: one statistical model (ARIMA), one classical ML model (Random Forest / XGBoost), and one **deep learning model (LSTM)** — all in scope and compared.
- **Interactive Streamlit dashboard/app** to visualize historical data, predictions, and model performance.
- Final report and presentation.

Out of Scope

- Real-time / streaming (intraday, tick-level) data prediction — daily granularity only.
- Alternative data sources (news sentiment, social media) — optional stretch goal only.
- Automated trade execution or brokerage integration.
- Guaranteeing or claiming investment performance.

4 · Target Users / Stakeholders

Stakeholder	Interest
Intern's Manager / Mentor	Evaluate intern's technical growth and deliverable quality
Data / Analytics Team	Potential reusable pipeline or internal reference project
Intern	Skill-building, portfolio project

5 · Data Requirements

Data Source

- Historical OHLCV data for the last 5 years for the Nifty 500 index, sourced from NSE and BSE historical data archives / APIs.

- Nifty 500 is the primary case study; individual constituent stocks may optionally be analyzed (e.g., top contributors to index movement) if time allows.
- NSE/BSE bhavcopy files or official historical downloads to be used as the authoritative source, ensuring accuracy and traceability.

Data Fields (Minimum)

- Date
- Open, High, Low, Close price
- Volume
- Adjusted Close (if available)

Data Quality Requirements

- No duplicate dates.
- Missing values (holidays, halted trading) handled via forward-fill or interpolation, clearly documented.
- Corporate actions (splits, dividends) accounted for via adjusted close where applicable.

6 · Functional Requirements

ID	Requirement
FR1	Ingest 5 years of historical OHLCV data for the Nifty 500 from NSE and BSE.
FR2	Clean and preprocess data (nulls, outliers, date alignment, NSE/BSE reconciliation).
FR3	Generate exploratory visualizations (price, volume, volatility, moving averages).
FR4	Engineer at least 5 technical / statistical features for modeling.
FR5	Train and evaluate 3 models: statistical (ARIMA), classical ML (RF / XGBoost), deep learning (LSTM).
FR6	Output predicted target price vs. actual, for backtesting and a future forecast horizon.
FR7	Present model performance metrics (RMSE, MAE, MAPE) with a full comparison table.
FR8	Provide an interactive Streamlit application to explore trends and compare model predictions.
FR9	Produce a final report summarizing findings, model comparison, and limitations.

7 · Non-Functional Requirements

- **Reproducibility:** code well-documented and runnable end-to-end (notebook or script).
- **Performance:** pipeline runs on a standard laptop / cloud notebook, no specialized infrastructure required.
- **Transparency:** assumptions, limitations, and data caveats clearly documented — especially market unpredictability.
- **Maintainability:** modular code (separate components for ingestion, cleaning, features, modeling, app).

8 · Methodology / Approach

Step	Phase	Description
1	Data Collection	Pull 5 years of Nifty 500 OHLCV data from NSE and BSE.
2	Data Cleaning	Handle missing values, align trading calendar, reconcile NSE/BSE discrepancies.
3	EDA	Visualize price trends, returns distribution, volatility clustering, volume correlation.
4	Feature Engineering	SMA, EMA, RSI, MACD, Bollinger Bands, lagged returns, rolling volatility.
5	Train / Test Split	Time-based split (last 6–12 months held out); no random shuffling.
6	Modeling	Naive baseline → ARIMA → Random Forest / XGBoost → LSTM (deep learning).
7	Evaluation	Compare all models on RMSE / MAE / MAPE against the target price.
8	Dashboard	Build Streamlit app for exploring data, models, and predicted vs. actual price.
9	Reporting	Summarize findings, visualize predictions, document limitations and next steps.

9 · Tools & Tech Stack

Category	Tools
Language	Python
Data Handling	pandas, numpy
Visualization	matplotlib, seaborn, plotly
Modeling	scikit-learn, statsmodels (ARIMA), XGBoost, TensorFlow/Keras or PyTorch (LSTM)
Data Source	NSE and BSE official historical data / bhavcopy downloads
Dashboard / App	Streamlit
Environment	Jupyter Notebook / Google Colab for development; Streamlit for the app
Version Control	GitHub

10 · Timeline

Suggested duration: **10 weeks**, extended from the original 6–8 week estimate to accommodate the added deep learning and Streamlit deployment scope.

Week	Milestone
1	Project setup, NSE/BSE data collection for Nifty 500, domain basics
2	Data cleaning & preprocessing
3	Exploratory Data Analysis (EDA)
4	Feature engineering

Week	Milestone
5	Baseline + statistical model (ARIMA)
6	Classical ML model(s) — Random Forest / XGBoost — build & tune
7	Deep learning model (LSTM) — build & tune
8	Model evaluation & comparison across all models
9	Build Streamlit dashboard / app
10	Final report, presentation, documentation, code cleanup

11 · Deliverables

- Cleaned Nifty 500 dataset (CSV), sourced from NSE/BSE, with documented preprocessing steps.
- EDA notebook / report with key visualizations.
- Feature-engineered dataset.
- Trained models — ARIMA, classical ML, and LSTM — with saved artifacts / code.
- Model evaluation summary with a full comparison table.
- **Deployed / runnable Streamlit app** for exploring data and target price predictions.
- Final presentation deck summarizing approach, findings, and limitations.
- Well-documented codebase (GitHub repo or shared folder).

12 · Risks & Limitations

Risk	Mitigation
Markets are highly noisy / near-random walk — accuracy may be low	Set realistic expectations; benchmark against naive baseline
Overfitting to historical patterns	Time-based train/test split, cross-validation, regularization
Data quality / availability issues from NSE/BSE sources	Validate data early; identify backup source
Corporate actions not accounted for	Use adjusted close prices; document known gaps
Misinterpretation as investment advice	Clear disclaimer: academic / learning exercise only

13 · Assumptions

- Data sourced from NSE and BSE official / historical channels; no paid third-party subscriptions required.
- Focus is on daily-level data, not intraday / high-frequency.
- Intern has, or will be given ramp-up time to learn, Python, pandas, Streamlit, and deep learning fundamentals (LSTM).
- The 5-year window ends at the most recent available trading date.

- "Target price" refers to a predicted future closing price at a defined forecast horizon (to be finalized).

14 · Decisions Confirmed

- ✓ Study subject: Nifty 500 index.
- ✓ Prediction target: target price (regression, not just directional classification).
- ✓ Data source: NSE and BSE.
- ✓ Delivery format: interactive Streamlit application.
- ✓ Deep learning (LSTM): in scope.

15 · Open Questions

- What is the exact forecast horizon for the target price — next day, next week, next month, or user-selectable in the app?
- Should NSE and BSE data be used together (cross-validated) or should one be primary and the other a backup?
- Should the Streamlit app be deployed for external access, or run locally for demo purposes only?
- Should individual Nifty 500 constituent stocks be analyzed, or should the project stay strictly at the index level?
- What compute resources are available for LSTM training — local machine vs. Colab / cloud GPU?

Disclaimer: *This project is intended for educational and analytical skill-building purposes only. Any predictions or insights generated should not be used as the basis for real investment decisions.*